

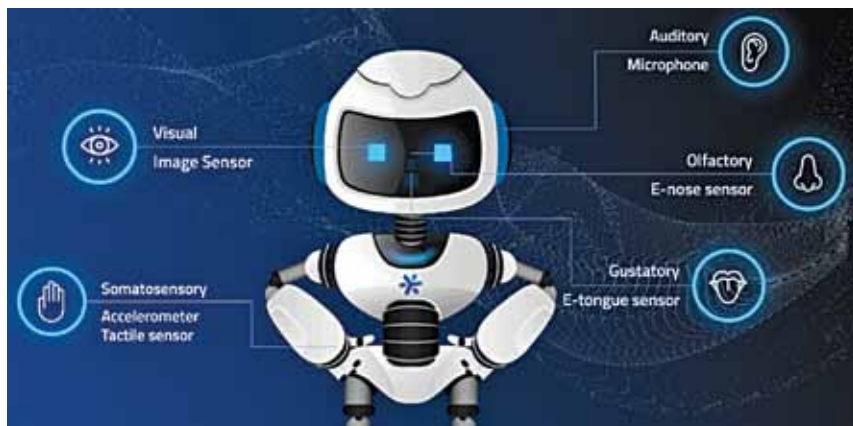
realised in the 1980s, starting with Carver Mead, an American scientist and engineer.

## Fast forward to today

Nowadays, almost all companies have dedicated teams working on neuromorphic computing. Ground-breaking research is being done in multiple research organisations and universities. It is safe to say that neuromorphic computing is gaining momentum and will continue to do so as various advancements are being made.

What's interesting to note is that although this is a specialised field with prerequisites from various topics, including solid-state physics, VLSI, neural networks, and computational neurobiology, undergraduate engineering students are extremely curious about this field.

At IIT Kanpur, Dr Shubham Sahay, Assistant Professor at the Department of Electrical Engineering, introduced a course on neuromorphic computing last year. Despite being a post-graduate level course, he saw great participation from undergrads as well. "Throughout the course, they were very interactive. The huge B.Tech participation in my course bears testimony to the fact that undergrads are really interested in this topic. I believe that this (neuromorphic computing) could be introduced as one of



The 5 sensor modalities (Source: BrainChip Inc.)

the core courses in the UG curriculum in the future," he says.

## Getting it commercial

Until recently, neuromorphic computing was a widely used term only in research and not in the commercial arena. However, as of January 18, 2022, BrainChip, a leading provider of ultra-low-power high-performance AI technology, commercialised its AKD1000 AIOT chip. Developers, system integrators, and engineers can now buy AKD1000-powered Mini PCIe boards and leverage them in their applications, especially those requiring on-edge computing, low power consumption, and high-performance AI.

"It's meant as our entry-level product. We want to proliferate this into as many hands as we can and get people designing in the Akida environment," says Rob Tel-

son, Vice President of WorldWide Sales at BrainChip. Anil Mankar, Co-founder and Chief Development Officer of BrainChip, explains, "We are enabling system integrators to easily use neuromorphic AI in their applications. In India, if some system integrators want to manufacture the board locally, they can take the bill of materials from us (BrainChip) and manufacture it locally."

What's fascinating about Akida is that it enables sensor nodes to compute without depending on the cloud. Further, BrainChip's AI technology not only performs audio and video based learning but even focuses on other sensor modalities like taste, vibration, and smell. You can use it to make a sensor that performs wine tasting! Here is a link to their wine tasting demonstration: <https://youtu.be/RwWTietRioM>

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